i.MX943 CRRM(Cyber-Resilient Recovery Module) API Rev 0.1

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Chapter 1

CRRM API

This document is a software reference description of the API provided by CRRM Application on i.MX943 Platform.

Chapter 2

Revision History

Revision	date	description
0.1	Aug 01 2025	Preliminary draft

Chapter 3

General concepts related to the API

3.1 CRRM : Cyber-Resilient Recovery Module

It enables early detection of attacks and adds the capability to recover to a trusted state without human intervention.

Chapter 4

Module Index

4.1 Modules

Here is a list of all modules:

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Chapter 5

Module Documentation

5.1 Boot Mode

Data Structures

- struct op_get_boot_mode_args_t
- struct op_set_boot_mode_args_t

Macros

- #define **NORMAL_BOOT_MODE** 0x69
- #define **RECOVERY_DOWNLOAD_BOOT_MODE** 0x3c
- #define **RECOVERY_INSTALL_BOOT_MODE** 0x5a

Functions

- crrm_err_t crrm_get_boot_mode (crrm_hdl_t crrm_hdl, op_get_boot_mode_args_t *args)
- crrm_err_t crrm_set_boot_mode (crrm_hdl_t crrm_hdl, op_set_boot_mode_args_t *args)

5.1.1 Detailed Description

5.1.2 Data Structure Documentation

5.1.2.1 struct op_get_boot_mode_args_t

Structure describing the get boot mode operation member arguments

Data Fields

uint8_t	boot_mode	Stores the boot mode Indicates the reason for the previous boot: • 0x69 = Normal (no CRRM events) • 0x3c = Recovery download (timeout of CRRM watchdog) • 0x5a = Recovery install (recovery image available) • Other values reserved
uint8_t	timer_id	If the current mode was entered due to timer expiry, provides the ID of the timer. Value 0xFF indicates no timer expired.

5.1.2.2 struct op_set_boot_mode_args_t

Structure describing the set boot mode operation member arguments

Data Fields

uint8_t	new_boot_mode	Stores the new boot mode to be set • 0x69 = Normal • 0x3c = Recovery download • 0x5a = Recovery install • All other values reserved
uint8_t	stored_boot_mode	Indicates the stored boot mode (may be different from requested mode if forbidden transition requested): • 0x69 = Normal • 0x3c = Recovery download • 0x5a = Recovery install • All other values reserved

5.1.3 Function Documentation

5.1.3.1 crrm_get_boot_mode()

```
crrm_err_t crrm_get_boot_mode (
            crrm_hdl_t crrm_hdl,
            op_get_boot_mode_args_t * args )
```

Perform get boot mode operation
User can call this function only after CRRM init has been done.

Get CRRM handle before calling this API.
This API request the boot mode information from the BBSM. For the case that the BBSM has just been initialized, the boot mode will be indicated as normal boot.

Parameters

crrm_hdl	handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code

5.1.3.2 crrm_set_boot_mode()

```
crrm_err_t crrm_set_boot_mode (
             crrm_hdl_t crrm_hdl,
             op_set_boot_mode_args_t * args )
```

Perform set boot mode operation
User can call this function only after CRRM init has been done.
Get CRRM handle before calling this API.
This API set the boot mode information in the BBSM. It is used by the recovery OS to request a mode transition. Only specific transitions are accepted, others are ignored. Accepted transitions are:
Recovery download -> recovery install (after recovery image successfully downloaded)
Recovery install -> normal (after recovery image successfully installed)
Recovery install -> recovery download (encountered error in recovery image install)
Only one transition can be requested per boot cycle.

Parameters

crrm_hdl	handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code

5.2 Get AWDT status

Data Structures

- struct AWDT_timer
- struct op_get_AWDT_status_args_t

Macros

- #define **TIMER_INFO_SIZE** 16
- #define **AWDT_TIMER_STATUS_SIZE** sizeof(AWDT_timer)

Functions

- crrm_err_t crrm_get_AWDT_status (crrm_hdl_t crrm_hdl, op_get_AWDT_status_args_t *args)

5.2.1 Detailed Description

5.2.2 Data Structure Documentation

5.2.2.1 struct AWDT_timer

Structure describing AWDT timer status parameters

Data Fields

uint8_t	initialized	0 = not initialized, 1 = initialized
uint8_t	running	0 = not running, 1 = running
uint16_t	BBSM_running	0 = not running, 1 = running
uint32_t	curr_timer_value	current timer value in ms
uint32_t	safety_period	safety period in ms
uint32_t	max_timer_value	max timer value in ms
uint32_t	BBSM_timer_value	stored timer value in ms
uint16_t	timer_info_len	length of timerInfo
uint8_t	timer_info[ ]	variable-length timerInfo data

5.2.2.2 struct op_get_AWDT_status_args_t

Structure describing the get status operation member arguments

Data Fields

uint8_t	timer_id	timer id
uint8_t *	addr	AWDT status report address.
uint32_t	input_size	AWDT status report size.
uint32_t	output_size	AWDT status report size provided by FW.
AWDT_timer *	status	status of timer with id timer_id

5.2.3 Function Documentation

5.2.3.1 crrm_get_AWDT_status()

```
crrm_err_t crrm_get_AWDT_status (
            crrm_hdl_t crrm_hdl,
            op_get_AWDT_status_args_t * args )
```

Perform get AWDT status operation
User can call this function only after CRRM init has been done.
Get CRRM handle before calling this API.
This API request a status report for a given timer.
The status data will be output to the memory address provided. If the buffer to copy this data is too small, no data will be copied.

Parameters

crrm_hdl	handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code

5.3 Get Nonce

Data Structures

- struct nonce_info
- struct op_get_nonce_args_t

Macros

- #define **NONCE_DATA_SIZE** 32

Functions

- crrm_err_t crrm_get_nonce (crrm_hdl_t crrm_hdl, op_get_nonce_args_t *args)

5.3.1 Detailed Description

5.3.2 Data Structure Documentation

5.3.2.1 struct nonce_info

Structure describing nonce info parameters

Data Fields

uint8_t	nonce_data[NONCE_DATA_SIZE]	nonce data
uint8_t	timer_info[ ]	variable-length timerInfo data

5.3.2.2 struct op_get_nonce_args_t

Structure describing the get nonce operation member arguments

Data Fields

uint8_t	timer_id	timer id
uint8_t *	nonce_addr	nonce info address
uint32_t	nonce_size	nonce info input size
uint32_t	output_size	nonce info size provided by FW
nonce_info *	info	nonce information (nonce data + timer info)

5.3.3 Function Documentation

5.3.3.1 crrm_get_nonce()

```
crrm_err_t crrm_get_nonce (
            crrm_hdl_t crrm_hdl,
            op_get_nonce_args_t * args )
```

Perform get nonce operation
User can call this function only after CRRM init has been done.
Get CRRM handle before calling this API.
This API request a new nonce value for a given timer.
This nonce value will invalidate any previous nonce values for the given timer.
The nonce plus programmed timerInfo will be output.
Note that if the buffer to copy this data is too small, no data will be copied but the nonce will still be updated inside the Sentinel.

Parameters

crrm_hdl	handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code

5.4 Handle

Typedefs

- typedef uint32_t crrm_hdl_t
- typedef hsm_err_t crrm_err_t

Functions

- crrm_hdl_t get_session_handle (void)

5.4.1 Detailed Description

5.4.2 Typedef Documentation

5.4.2.1 crrm_hdl_t

```
typedef uint32_t crrm_hdl_t
```

Define the CRRM handle type

5.4.2.2 crrm_err_t

```
typedef hsm_err_t crrm_err_t
```

Define the CRRM error type. Please refer ELE API guide for error codes.

5.4.3 Function Documentation

5.4.3.1 get_session_handle()

```
crrm_hdl_t get_session_handle (
            void  )
```

Return CRRM handle to be used while requesting CRRM services
Check if you have received a valid (non-zero) session handle before continuing.
This is the first API to be called.

Returns

CRRM handle

5.5 Refresh AWDT

Data Structures

- struct signature_data
- struct request_data
- struct op_refresh_AWDT_args_t

Macros

- #define **NONCE_DATA_SIZE** 32

Functions

- crrm_err_t crrm_refresh_AWDT (crrm_hdl_t crrm_hdl, op_refresh_AWDT_args_t *args)

5.5.1 Detailed Description

5.5.2 Data Structure Documentation

5.5.2.1 struct signature_data

Structure describing signature data parameters to be signed

Data Fields

uint8_t	timer_id	timer id
uint8_t	pub_key_index	public key index
uint16_t	reserved	reserved value
uint32_t	new_timer_value	new value for timer
uint8_t	nonce_data[NONCE_DATA_SIZE]	nonce data
uint8_t	timer_info[ ]	variable-length timerInfo data

5.5.2.2 struct request_data

Structure describing request data parameters

Data Fields

uint32_t	new_timer_value	new value for timer
uint16_t	signature_length	length of signature field
uint8_t	signature[ ]	Signature calculated on the signature data.

5.5.2.3 struct op_refresh_AWDT_args_t

Structure describing the refresh AWDT operation member arguments

Data Fields

uint8_t	timer_id	timer id
uint8_t	pub_key_index	public key index
uint8_t *	req_data_addr	nonce info address
uint32_t	req_data_size	nonce info input size
uint32_t	new_timer_value	new value for timer

5.5.3 Function Documentation

5.5.3.1 crrm_refresh_AWDT()

```
crrm_err_t crrm_refresh_AWDT (
            crrm_hdl_t crrm_hdl,
            op_refresh_AWDT_args_t * args )
```

Perform refresh AWDT operation
User can call this function only after CRRM init has been done.
Get CRRM handle before calling this API.
This API is used to refresh the specified AWDT.
The message contains a signature, that is calculated over the concatenation of the following information:
1: word 1 of the message below (pub key index and timer ID)
2: the newTimerValue from the request structure
3: current nonce value for the requested timer
4: timer info for the requested timer if present
If the signature is valid, the corresponding AWDT timer is refreshed

Parameters

crrm_hdl	handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code

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